Gree...
AP, Dept of EEE

Measurement of resistance

Classification of Resistance

From the point of view of measurements, resistances can be classified as follows:

i. <u>Low resistances</u>:- All resistances of the <u>order of 1Ω</u> and under may be classified as low resistances. (upto 1Ω) Measurement of low resistances is required for determination of resistance of armature and series field windings of large machines, ammeter shunts, cable lengths, contacts etc.

ii <u>Medium resistances</u>:- This class includes resistances from about <u>1Ω upwards</u> to about <u>100kΩ</u>. Most of the electrical apparatus employed are of medium resistance.

iii. <u>High resistances</u>:- Resistances of the order of <u>100 kΩ</u> and above are classified as high resistances.

Measurement of Low Resistances

Following methods are used for the measurement of low resistances:

1. Ammeter - Voltmeter method
2. Potentiometer method
3. Kelvin double bridge method

$V = IR$

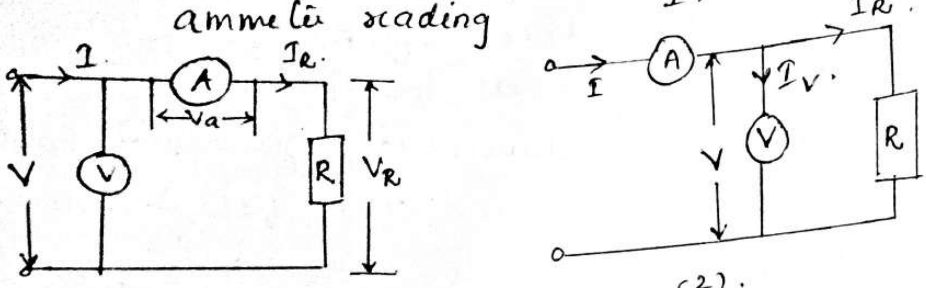
$R = \frac{V}{I}$

1. <u>Ammeter - Voltmeter method</u>

The resistance is nothing but the ratio of voltage and current. Thus measurement of voltage & current separately is sufficient for the resistance measurement. Thus voltmeter - ammeter method is based on the principle of measuring voltage and current separately. This is very simple method as the instruments required to measure voltage & current are readily available in laboratory.

Two types of connections are employed for ammeter - voltmeter method. are shown in fig: (1) & (2). In the both cases if readings of ammeter and voltmeter are taken, then the measured value of resistance (R_m) is given by;

$$R_m = \frac{\text{voltmeter reading}}{\text{ammeter reading}} = \frac{V}{I}.$$



(1) (2).

fig: Measurement of resistance by ammeter - voltmeter method

∴ The measured value of resistance, R_m = true value, R.

$$R_m = R.$$

Consider circuit of fig: (1)

In this ckt ammeter measures the true value of the current through the resistance but the voltmeter does not measure the true voltage across the resistance

» The voltmeter indicates the sum of the voltages across the ammeter and the measured resistance.

Resistance of Ammeter → R_a.

Voltage across ammeter → $V_a = I R_a$.

R_{m1} → Measured value of resistance.

$$R_{m1} = \frac{V}{I} = \frac{V_R + V_a}{I} = \frac{IR + IR_a}{I} = R + $$

$$\boxed{R_{m1} = R + R_a}$$

∴ True value of resistance $R = R_{m1} - R_a$.

⇒ $$R = R_{m1}\left[1 - \frac{R_a}{R_{m1}}\right]$$

Thus, the measured value of resistance is higher than the true value $R = R_{m1} - R_a$.

It is also clear from above eqn: that the true value is equal to the measured $(R = R_m)$ only if the ammeter resistance R_a is zero $[R_a = 0]$

$$R = R_{m1} - R_a \qquad R_a = 0$$

$$\underline{\underline{R = R_{m1}}}$$

Consider the circuit of fig: (2)

In this ckt the voltmeter measures the true value of voltage but the ammeter measures the sum of currents through the resistance of the voltmeter.

Current through the voltmeter $I_V = \frac{V}{R_V}$.

Measured value of resistance, $R_{m2} = \frac{V}{I} = \frac{V}{I_R + I_V} = \frac{V}{\frac{V}{R} + \frac{V}{R_V}}$

$$= \frac{1}{\frac{1}{R} + \frac{1}{R_V}} = \frac{1}{\frac{1}{R}\left[1 + \frac{R}{R_V}\right]}$$

$$R_{m2} = \frac{R}{1 + \frac{R}{R_V}} = \frac{R}{\frac{R_V + R}{R_V}}$$

True value of resistance $R = \frac{R_{m2} R_V}{R_V - R_{m2}} = R_{m2} \times \frac{R_V}{R_V\left(1 - \frac{R_{m2}}{R_V}\right)}$

$$R = R_{m2}\left[\frac{1}{1 - \frac{R_{m2}}{R_V}}\right] \quad \text{—— (1)}$$

from above eqn:
it is clear that the true value of resistance is equal to the measured value only if the resistance of voltmeter, R_V is infinite.

3. ## Wheatstone Bridge.

The simplest form of bridge is for the purpose of measuring resistance and is called the Wheatstone bridge.

Basic operation.

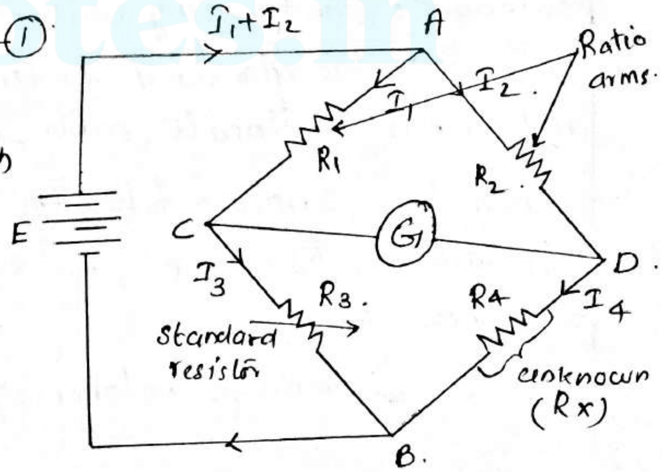
» The bridge has four resistive arms, together with a source of emf (a battery) and null detector, usually a galvanometer (G) is used as a null detector.

» The current through the galvanometer depends on the potential difference between the points C & D.

» The bridge is said to be balanced when the potential difference across the galvanometer is zero volt (0V) so that there is no current through the galvanometer.

• Hence the bridge is balanced when potential difference between points C & D is equal,

ie $I_1 R_1 = I_2 R_2$ —— (1)



• When the current through the galvanometer is zero; the following conditions should be satisfied.

$$I_1 = I_3 = \frac{E}{R_1+R_3} \quad \text{—— (2)}$$

and $$I_2 = I_4 = \frac{E}{R_2+R_4} \quad \text{—— (3)}$$

sub: the value of I_1 & I_2 in eqn: (1), we get.

$$\frac{E}{R_1+R_3} \times R_1 = \frac{E}{R_2+R_4} \times R_2$$

$$\frac{R_1}{R_1+R_3} = \frac{R_2}{R_2+R_4} \quad \text{—— (4)}$$

or $R_1(R_2+R_4) = R_2(R_1+R_3)$.

$$R_1R_2 + R_1R_4 = R_1R_2 + R_2R_3.$$

$$R_1R_4 = R_2R_3 \quad \text{——— (5)}$$

eqn: (5) is the well known expression for balance of the Wheatstone bridge.

>> When the 3 of the resistors are known, the 4th may be determined. Thus if R_4 is the unknown resistor, its resistance R_x can be found as follows:

$$R_x = \frac{R_2}{R_1} \cdot R_3.$$

>> The resistors R_2 & R_1 are called ratio arms while the resistor R_3 is called the standard arm of the bridge.

Applications of Wheatstone Bridge.

1. Measurement of the resistance of motor windings, transformers, solenoids and relay coils.
2. This bridge is also used extensively by telephone companies and others to locate cable faults.

1. Above fig: consists of following parameters
$R_1 = 20k\Omega$, $R_2 = 30k\Omega$, $R_3 = 80k\Omega$. Determine the unknown resistance R_x.

Using Bridge balance eqn: $\Rightarrow R_1R_x = R_2R_3$.

$$R_x = \frac{R_2R_3}{R_1} = \frac{30 \times 80}{20}$$

$$= \underline{\underline{120k\Omega}}.$$

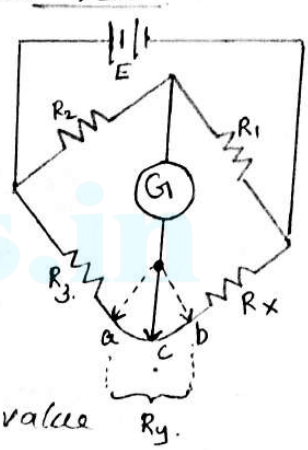
Kelvin double bridge method

Firstly we shall discuss kelvin bridge & thereafter Kelvin double bridge method will be described in detail.

Kelvin bridge.

>> modified version of Wheatstone bridge.

>> provides greatly increased accuracy in the measurement of low-value resistances, generally below 1Ω.



- R_y represents the resistance of connecting lead from R_3 to R_x.

- Two galvanometer connections are available to point a or to point b

>> When the galvanometer is connected to point 'a', the resistance R_y is added to the unknown resistor R_x. hence the value measured by the bridge, indicates much higher value of R_x. $R_y + R_x$

>> If the galvanometer is connected to terminal 'b', then R_y gets added to R_3. This results in the measurement of R_x much lower than the actual value. $R_y + R_3$

>> When the galvanometer connection is made to point c in between the two points a & b; in such a way

That the ratio of the resistances from b to c & from a to c equals the ratio of resistors R_1 & R_2, then

$$\frac{R_{bc}}{R_{ac}} = \frac{R_1}{R_2} \quad \text{——} (1)$$

The balance eqn: for the bridge in its standard form is.

$$R_1 R_3 = R_2 R_X. \quad \text{——} (2)$$

But R_3 & R_X now are changed to $R_3 + R_{ac}$ and $R_X + R_{bc}$ respectively due to lead resistance. $R_3 = R_3 + R_{ac}$

$R_X = R_X + R_{bc}$

$$R_1 (R_3 + R_{ac}) = R_2 (R_X + R_{bc}).$$

$$(R_X + R_{bc}) = \frac{R_1}{R_2} (R_3 + R_{ac}). \quad \text{——} (3)$$

Now we have; $\dfrac{R_{bc}}{R_{ac}} = \dfrac{R_1}{R_2}$.

$$\frac{R_{bc}}{R_{ac}} + 1 = \frac{R_1}{R_2} + 1 \rightarrow \text{Adding 1 to both sides.}$$

$\longrightarrow (4)$

$$\frac{R_{bc} + R_{ac}}{R_{ac}} = \frac{R_1 + R_2}{R_2}. \quad \text{——} (5)$$

But $R_{bc} + R_{ac} = R_y$.

sub: in eqn:(5) we get;

$$\frac{R_y}{R_{ac}} = \frac{R_1 + R_2}{R_2}. \quad \text{——} (6)$$

$$R_{ac} = \frac{R_2 R_y}{R_1 + R_2}. \quad \text{——} (7)$$

Now; $R_{bc} + R_{ac} = R_y$.

$$R_{bc} = R_y - R_{ac} \quad \text{——} (8)$$

sub: eqn:(7) into eqn: (8)

sub: eqn: (7) in eqn:(8)

$$R_{bc} = R_y - \frac{R_2 R_y}{R_1 + R_2} = R_y\left[1 - \frac{R_2}{R_1+R_2}\right].$$

$$R_{bc} = \frac{R_1 R_y + \cancel{R_2 R_y} - \cancel{R_2 R_y}}{(R_1+R_2)} = \frac{R_1 R_y}{R_1 + R_2}.$$

$$R_{bc} = \frac{R_1 R_y}{R_1 + R_2} \quad \text{——} (9).$$

substituting for R_{bc} & R_{ac} in eqn: (3) we get;

$$(3) \Rightarrow R_x + R_{bc} = \frac{R_1}{R_2}\left(R_3 + R_{ac}\right).$$

$$R_x + \frac{R_1 R_y}{R_1 + R_2} = \frac{R_1}{R_2}\left(R_3 + \frac{R_2 R_y}{R_1+R_2}\right).$$

$$R_x + \cancel{\frac{R_1 R_y}{R_1 + R_2}} = \frac{R_1 R_3}{R_2} + \cancel{\frac{R_1 R_y}{R_1 + R_2}}.$$

$$\boxed{R_x = \frac{R_1 R_3}{R_2}} \quad \text{——} (10)$$

Thus eqn: (10) represents standard bridge balance eqn: for the Wheatstone bridge. Thus the effect of the connecting lead resistance is completely eliminated by connecting the galvanometer to an intermediate position 'c'.

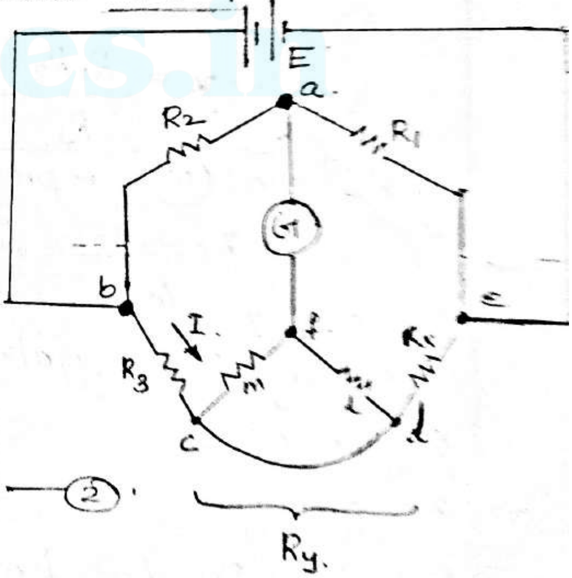
→ This principle forms the basis of the construction of Kelvin's Double Bridge which is popularly called Kelvin Bridge.

Kelvin Double Bridge for Measurement of low resistance.

>> This bridge consist of another set of ratio arms hence called double bridge.

>> The second set of ratio arms is the resistances 'l' and 'm'. This second set of ratio arms connects the galvanometer to a point 'f' at the appropiate potential between 'c' & 'd' and it eliminates the effect of the yoke resistance 'R_y'. The ratio of the resistances of arms 'l' & 'm' is the same as the ratio of R_1 & R_2 $\Rightarrow$ ie $\frac{R_1}{R_2} = \frac{l}{m}$.

>> The galvanometer will read zero (null indication) when potential at 'a' = potential at 'f'.



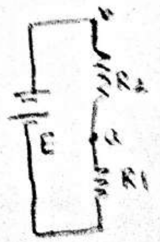
ie, $E_{ab} = E_{bcf}$ (by voltage division rule)

But $E_{ab} = \frac{R_2}{R_1+R_2} \times E$ —— (1)

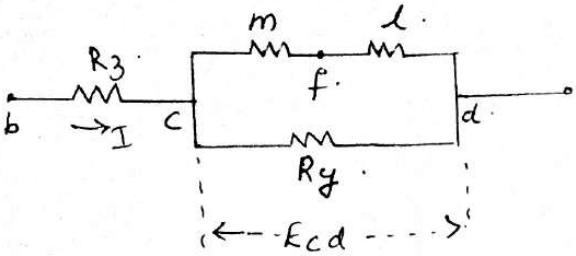
and $E = I\left[R_3 + R_x + \frac{(l+m)R_y}{(l+m)+R_y}\right]$ —— (2)

Sub: for E in eqn: (1), we get

$$E_{ab} = \frac{R_2}{R_1+R_2} \times I\left[R_3 + R_x + \frac{(l+m)R_y}{(l+m)+R_y}\right]$$



For E_{bcf}, consider the path from the terminal b to f as shown in fig. below.



$$E_{bcf} = IR_3 + E_{cf}$$

$$E_{cd} = I\left[\frac{R_y\,(l+m)}{R_y+l+m}\right] \quad \text{and} \quad E_{cf} = \frac{m}{l+m}E_{cd}$$

(Sub for Ecd)

$$E_{cf} = \frac{m}{l+m} \times I\left[\frac{R_y\,(l+m)}{R_y+l+m}\right]$$

$$E_{bcf} = IR_3 + E_{cf} = IR_3 + \frac{m}{l+m} \cdot I\left[\frac{R_y(l+m)}{R_y+l+m}\right]$$

$$E_{bcf} = I\left[R_3 + \frac{m}{l+m}\,\frac{R_y\,(l+m)}{R_y+(l+m)}\right]$$

Now $E_{ab} = E_{bcf}$; for balancing.

$$\frac{IR_2}{R_1+R_2}\left(R_3 + R_x + \frac{(l+m)\,R_y}{(l+m)+R_y}\right) = I\left[R_3 + \frac{m}{l+m}\left\{\frac{(l+m)\,R_y}{(l+m)+R_y}\right\}\right]$$

(or); $$R_3 + R_x + \frac{(l+m)\,R_y}{(l+m)+R_y} = \frac{R_1+R_2}{R_2}\left(R_3 + \frac{m\,R_y}{(l+m)+R_y}\right)$$

$$R_3 + R_x + \frac{(l+m)\,R_y}{(l+m)+R_y} = \left[\frac{R_1}{R_2}+1\right]\left(R_3 + \frac{m\,R_y}{l+m+R_y}\right)$$

$$R_x + \frac{(l+m)\,R_y}{(l+m)+R_y} + \cancel{R_3} = \frac{R_1R_3}{R_2} + \cancel{R_3} + \frac{m\,R_1R_y}{R_2\,(l+m+R_y)} + \frac{m\,R_y}{l+m+R_y}$$

$$R_x = \frac{R_1R_3}{R_2} + \frac{m\,R_1R_y}{R_2(l+m+R_y)} + \frac{m\,R_y}{l+m+R_y} - \frac{(l+m)\,R_y}{l+m+R_y}$$

$$R_x = \frac{R_1 R_3}{R_2} + \frac{m R_1 R_y}{R_2(l+m+R_y)} + \frac{m R_y - l R_y - m R_y}{l+m+R_y}.$$

$$R_x = \frac{R_1 R_3}{R_2} + \frac{m R_1 R_y}{R_2(l+m+R_y)} - \frac{l R_y}{l+m+R_y}$$

$$\boxed{R_x = \frac{R_1 R_3}{R_2} + \frac{m R_y}{l+m+R_y}\left[\frac{R_1}{R_2} - \frac{l}{m}\right]}$$

But

$$\frac{R_1}{R_2} = \frac{l}{m}.$$

$$R_x = \frac{R_1 R_3}{R_2} + \frac{m R_y}{l+m+R_y}\left(\frac{R_1}{R_2} - \frac{R_1}{R_2}\right).$$

$$\boxed{R_x = \frac{R_1 R_3}{R_2}}$$

This is the standard equation of the bridge balance. The resistances l, m & R_y are not present in the eqn. Thus the effect of lead & contact resistance is completely eliminated.

→ In a typical kelvin's double bridge, the range of a resistance covered is 1Ω to 10MΩ with an accuracy of ± 0.05% to ± 0.2%

1. In a kelvin double bridge, there is error due to mismatch between the ratios of outer and inner arm resistances. The following data relate to this bridge. Standard resistance = 100.03 μΩ, Inner ratio arms = 100.31 Ω & 200Ω, Outer ratio arms = 100.24Ω and 200Ω. The resistance of connecting leads from standard to unknown resistor is 680 μΩ. Calculate the unknown resistance?

Q2 Given $R_3 = 100.03$, $l = 100.31\Omega$, $m = 200\Omega$, $R_1 = 100.24\Omega$

$R_2 = 200\Omega$, $R_y = 680\mu\Omega$.

MEASUREMENT OF INSULATION RESISTANCE

Resistances can be classified into

i) Low resistance - upto 1Ω

ii) Medium resistance - 1Ω to $100k\Omega$

iii) high resistance - greater than $100k\Omega$

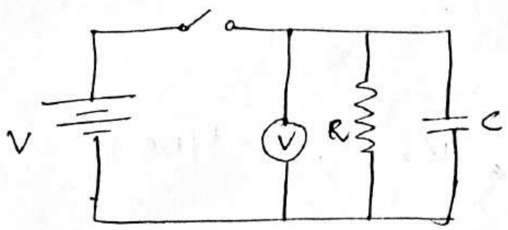
Low resistance can be measured by Ammeter voltmeter method, potentiometer method etc.

Medium resistance can be measured by wheatstone bridge, Ammeter voltmeter method etc.

For measurement of high resistances such as resistance of insulation, we use method known as Loss of charge method.

Loss of charge method:-

In this method, the insulation resistance R to be measured is connected in parallel with a capacitor c and a electrostatic voltmeter.



The capacitor is charged to some suitable voltage by means of a battery having voltage V and is then allowed to discharge through a resistance. The terminal voltage is observed over a considerable period of time during discharge.

The voltage across the capacitor at any instant after the application of voltage is

$$V = v\,e^{(-t/RC)}$$

$$\Rightarrow \frac{V}{v} = e^{-t/RC}$$

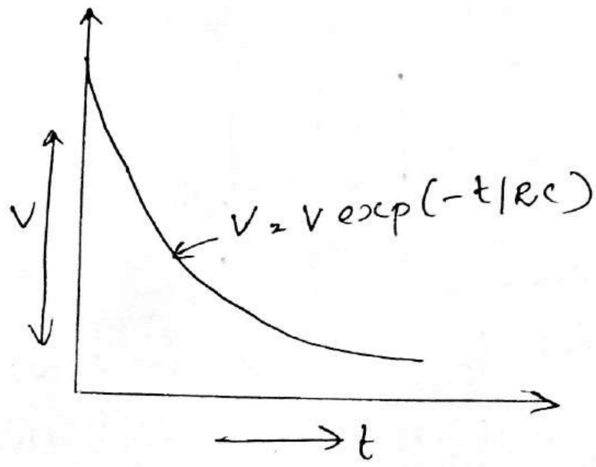
Taking natural logarithm on both sides

$$\Rightarrow \ln\left(\frac{V}{v}\right) = \frac{-t}{RC}$$

$\therefore$ Insulation resistance $R = \dfrac{t}{C\ln(V/v)}$.

$$\boxed{R = \frac{0.4343\,t}{C\log\left(\frac{V}{v}\right)}} \quad \text{—— (1)}$$

The variation of voltage with time is shown below,



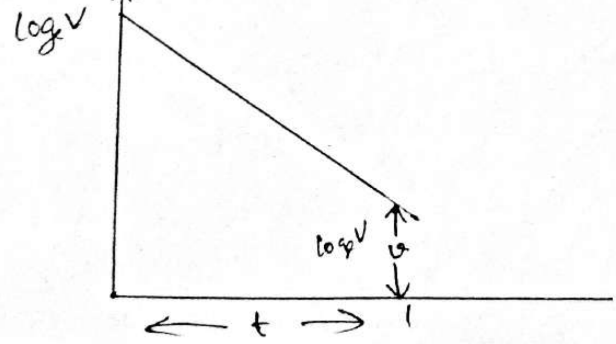
From ①, it is clear that if V, v, c and t are known, value of R can be computed.

If resistance R is very large the time for an appreciable fall in voltage is very large and thus this process may become time consuming. Also the voltage time curve will thus be very flat and unless great care is taken in measuring voltage at the beginning and end of the time t, a serious error may be made in the ratio V/v

More accurate results can be obtained by change in the voltage V-v directly & calling this change 'e'

$$\therefore R = \frac{0.4343\, t}{c \log_{10} \frac{V}{V-e}}$$

It may be advisable to determine time t from the discharge curve of the capacitor by plotting curve of $\log_e v$ against time t.



Methods to measure earth resistance are,

1) Fall of potential Method:-

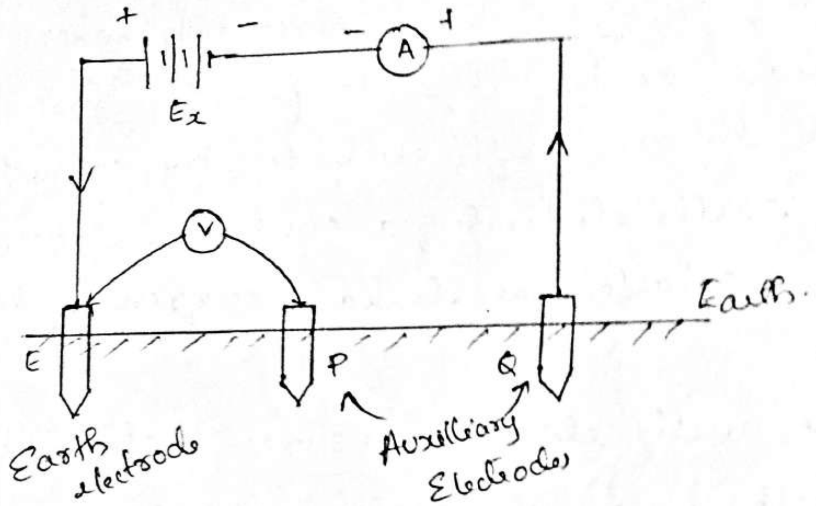
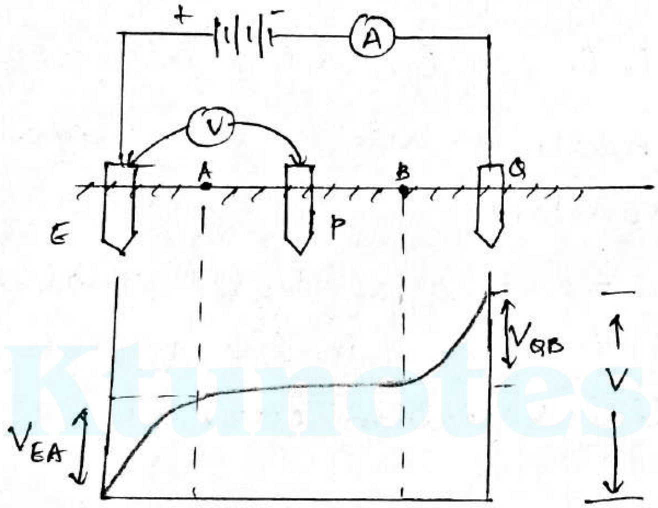


Figure shows the circuit diagram for measurement of earth resistance by fall of potential method E is the earth electrode The electrode Q is the auxilliary electrode. A current I is passed through the electrodes E and Q with the help of external battery

Another auxillary electrode P is introduced in between the electrodes E and Q. The voltage between the electrodes E and P is measured with the help of voltmeter. Let this voltage be V for the known current I

The current density near the electrode E & Q will be high.

Thus if the distance of electrode P is changed from electrode E and electrode Q, the electrode P experiences a change in potential near the electrode while a constant potential between the electrode but away from electrode. Such a potential variation between the electrodes E and Q is shown below.



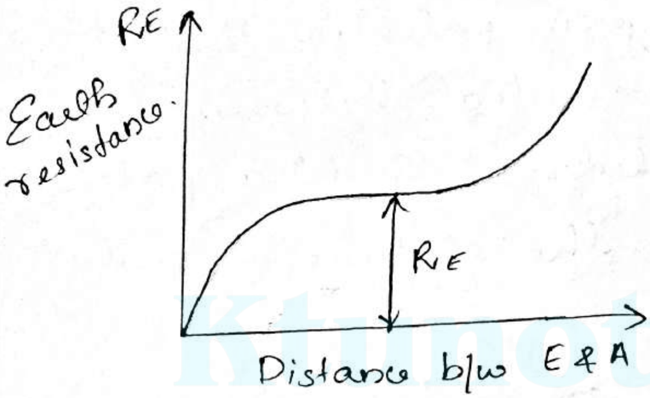
The potential rises near the electrodes E and Q due to higher current density in the proximity of the electrodes. while in the middle section potential is constant.

By measuring the potential between the electrodes E & P as V_{EP}, earth resistance can be obtained as

$$R_E = \frac{V_{EP}}{I}$$

The measurement of V_{EP} is done at many points by changing position of electrode P

The graph is plotted which shows the variation of R_E against the distance between E & P.



· Near the electrodes E and Q earth resistance increases rapidly while in the middle section AB it remains constant.

The true reading of R_E is that when the measurement is taken in the middle section A-B where potential variation curve is absolutely flat avoiding ...

Earth tester.

The construction of the megger is discussed in the last section. There are some modifications done in the basic megger to design a device called megger earth tester. It has some additional constructional features. they are: (i) A rotating current reverser.
(ii) A rectifier

Both these parts have L type commutators and both are mounted on the shaft of the generator which is to be driven manually, with the help of handle provided to it. Each commutator has four fixed brushes. Two brushes of each commutator are so placed that they make contact alternately with each segment commutator, as commutator rotates. While other two brushes of each commutator are placed in such a way that they make contact always with only one segment of the commutator, irrespective of the commutator position.

The earth tester has four terminals P_1 P_2 & C_1 C_2 taken out from two brushes of each commutator. The arrangement of earth tester is shown in fig. below.

The terminals P_1 & C_1 are shorted using a link & common point is connected to earth electrode E. The other two terminals P_2 & C_2 are connected to auxillary electrodes P & Q respectively.

The ratio of the voltage sensed by voltage coil & current through current coil, decides the deflection of the pointer. These quantities are proportional to the current I & the voltage between the electrodes E & P. Hence the deflection of the pointer directly gives the reading of the earth resistance R_E. Basically earth tester is permanent magnet moving coil device which can be used for d.c purposes only. But to have the facility for the measurements with alternating currents flowing in the earth, current reverser & rectifier are provided. Generally measurements with a.c is preferred as flow of a.c current through soil has few advantages such as the elimination of effects corresponding to production of back emf in the soil due to presence of electrolytic action in the soil.

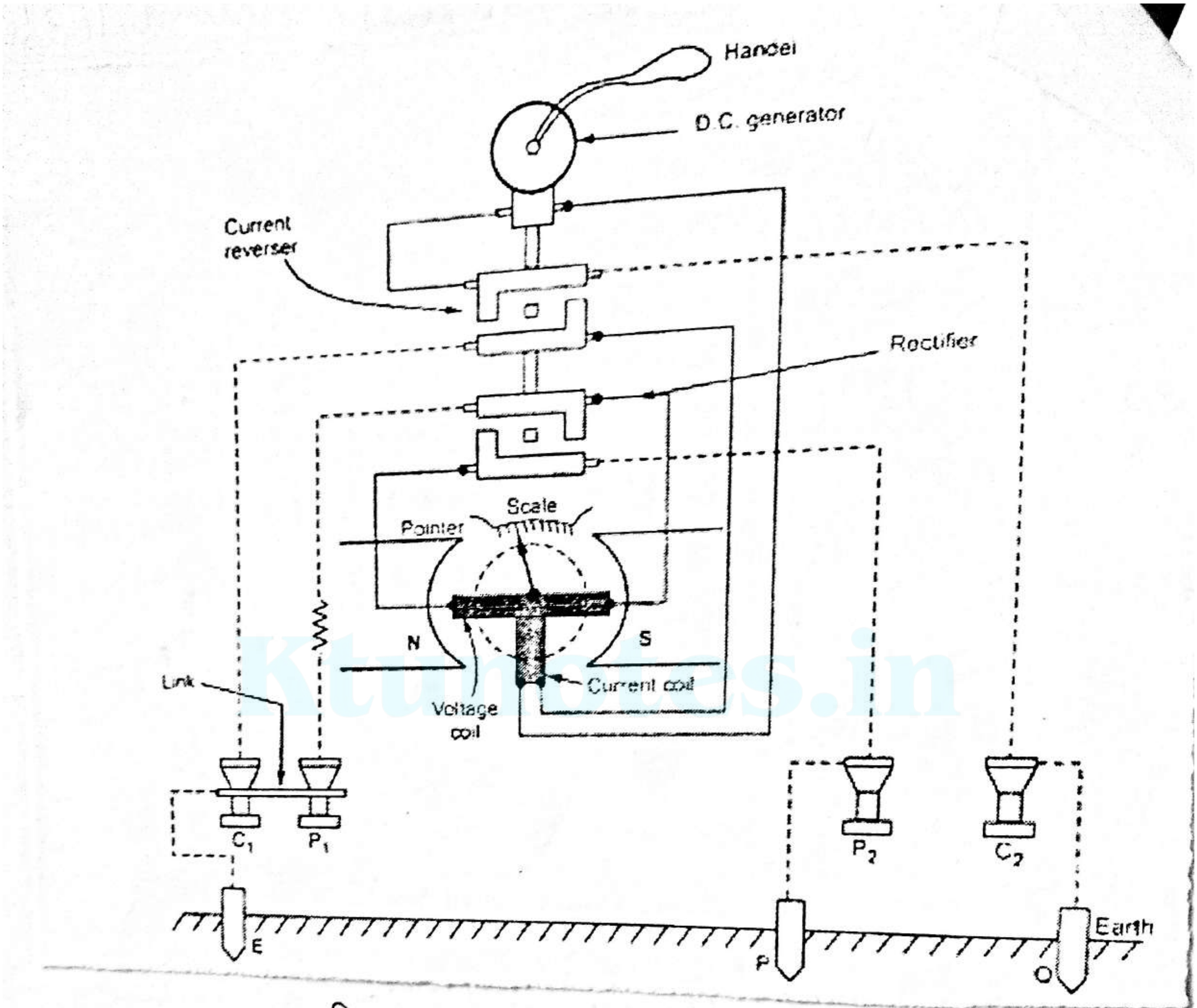


fig: Earth tester.

for frequency ω = 7500 radians/sec., find the shunt-less resistance and capacitance of the imperfect condenser. (UPTU, 200...)

Solution. Refer to Fig. 6.23 : *Given* : $\omega = 7500$ rad/s; $R_2 = 140\ \Omega$; $R_3 = R_4 = 1000\ \Omega$; $C_2 = 0.0115\ \mu$ F.

We know that for balance,

$$\bar{Z}_1 \bar{Z}_4 = \bar{Z}_2 \bar{Z}_3 \quad ...(i)$$

where,

$$\bar{Z}_1 = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega C_1 R_1}{j\omega C_1}$$

$$\bar{Z}_2 = R_2 + \frac{1}{j\omega C_2} = \frac{1 + j\omega C_2 R_2}{j\omega C_2}$$

$$\bar{Z}_3 = R_3$$

$$\bar{Z}_4 = R_4$$

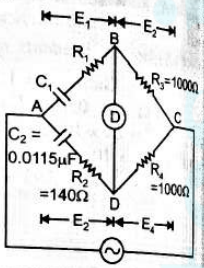


Fig. 6.23.

Inserting the values in eqn.(*i*), we get

$$\left(\frac{1 + j\omega C_1 R_1}{j\omega C_1}\right) R_4 = \left(\frac{1 + j\omega C_2 R_2}{j\omega C_2}\right) R_3$$

or,

$$j\omega C_2 R_4 = \omega^2 R_1 C_1 C_2 R_4$$
$$= j\omega C_1 R_3 - \omega^2 C_1 C_2 R_2 R_3$$

Comparing the *real part*, we have :

$$\omega^2 R_1 C_1 C_2 R_4 = \omega^2 C_1 C_2 R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4} \quad ...(ii)$$

or,

$$R_1 = \frac{140 \times 1000}{1000} = \mathbf{140\ \Omega\ (Ans.)}$$

Comparing the *imaginary part*, we get :

$$\omega C_2 R_4 = \omega C_1 R_3$$

or,

$$C_2 R_4 = C_1 R_3 \quad ...(iii)$$

or,

$$C_1 = \frac{C_2 R_4}{R_3} = \frac{(0.0115 \times 10^{-6}) \times 1000}{1000}$$
$$= 0.0115 \times 10^{-6}\ \text{F} \quad \text{or} \quad \mathbf{0.0115\ \mu F\ (Ans.)}$$

MEASUREMENT OF INDUCTANCE

6.3.2. Maxwell Bridge (*for measurement of "inductance"*)

Maxwell bridge, shown in Fig. 6.24 *measures an unknown inductance in terms of a known capacitor* :

— The use of standard arm offers the advantage of compactness and easy shielding.

— The capacitor is almost a loss-less component.

— One arm has a resistance R_1 in parallel with C_1, and hence it is easier to write the balance equation using the admittance of arm 1 instead of the impedance.

The general equation for bridge balance is,

$$\bar{Z}_1 \bar{Z}_x = \bar{Z}_2 \bar{Z}_3$$

or,

$$\bar{Z}_x = \frac{\bar{Z}_2 \bar{Z}_3}{\bar{Z}_1} = \bar{Z}_2 \bar{Z}_3 \bar{Y}_1 \quad ...(6.57)$$

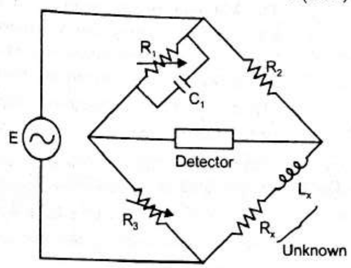


Fig. 6.24. Maxwell bridge for inductance measurements.

where,

$$\bar{Z}_1 = R_1$$

in parallel with C_1 *i.e.* $\bar{Y}_1 = \frac{1}{\bar{Z}_1}$

$$\bar{Y}_1 = \frac{1}{R_1} + j\omega C_1$$

$$\bar{Z}_2 = R_2$$

$$\bar{Z}_3 = R_3$$

$$\bar{Z}_x = R_x \text{ in series with } L_x$$

$(= R_x + j\omega L_x)$

Substituting these values in eqn. (6.57), we get

$$R_x + j\omega L_x = R_2 R_3 \left(\frac{1}{R_1} + j\omega C_1\right)$$

or,

$$R_x + j\omega L_x = \frac{R_2 R_3}{R_1} + j\omega C_1 R_2 R_3$$

Equating *real terms* and *imaginary terms*, we have

$$R_x = \frac{R_2 R_3}{R_1}; \quad ...(6.58)$$

$$L_x = C_1 R_2 R_3$$

To obtain the bridge balance, first R_3 is adjusted for inductive balance and R_1 is adjusted for resistive balance.

The *"quality factor"* of the coil is given by,

$$Q = \frac{\omega L_x}{R_x} = \frac{\omega . C_1 R_2 R_3}{(R_2 R_3 / R_1)} = \frac{\omega . C . R_2 R_3 R_1}{R_2 R_3}$$

i.e.

$$Q = \omega C_1 R_1 \quad ...(6.59)$$

- Maxwell bridge is limited to the measurement of *low Q values (1-10)*.
- The bridge is *particularly suited for inductance measurements*, since comparison with a capacitor is more ideal than with another inductance. Commercial bridges measure *from 1-1000 H, with ± 2 % error* (If Q is very large, R_1 becomes excessively large and it is impractical to obtain a satisfactory variable standard resistance in the range of values required).

Advantages :

1. The balance equation is independent of losses associated with inductance.
2. The measurement is independent of the excitation *frequency*.
3. The scale of the resistance can be calibrated to read inductance directly.
4. A wide range of inductance at power and audio frequencies can be measured.

6.3.7. Schering Bridge (*for Measurement of* "Capacitance")

It is one of the most important A.C. bridges, widely used for the *measurement of unknown capacitors, dielectric loss and power factor.*

Fig. 6.34 shows the basic circuit arrangement of Schering bridge. The capacitor C_3 is a high quality mica capacitor (low-loss) for general measurements, or an air capacitor (having a very stable value and a very small electric field) for insulation measurement.

The general equation for balance is,

$$\bar{Z}_1 \bar{Z}_x = \bar{Z}_2 \bar{Z}_3$$

or,
$$\bar{Z}_x = \frac{\bar{Z}_2 \bar{Z}_3}{\bar{Z}_1} = \bar{Z}_2 \bar{Z}_3 \bar{Y}_1 \qquad ...(i)$$

where,
$$\bar{Z}_x = R_x - \frac{j}{\omega C_x}$$

$$\bar{Z}_2 = R_2$$

$$\bar{Z}_3 = -\frac{j}{\omega C_3}$$

$$\bar{Y}_1 = \frac{1}{R_1} + j\omega C_1$$

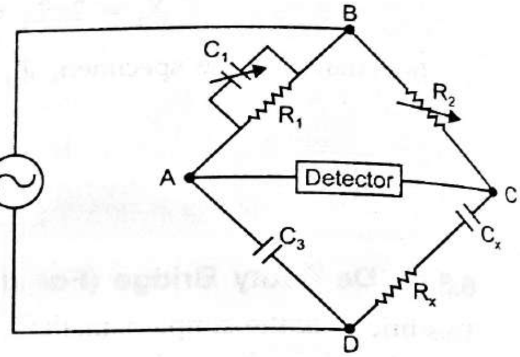


Fig. 6.34. Schering bridge for the measurement of capacitance.

Substituting the values in eqn. (*i*), we get

$$R_x - \frac{j}{\omega C_x} = R_2\left(-\frac{j}{\omega C_3}\right) \times \left(\frac{1}{R_1} + j\,\omega\,C_1\right)$$

or
$$\left(R_x - \frac{j}{\omega C_x}\right) = \frac{R_x\,(-j)}{R_1\,\omega\,C_3} + \frac{R_2\,C_1}{C_3}$$

Equating the *real* and *imaginary* terms, we get :

$$R_x = \frac{R_2\,C_1}{C_3} \qquad ...(6.70)$$

and,
$$C_x = \frac{R_1\,C_3}{R_2} \qquad ...(6.71)$$

The dial of capacitor C_1 can be calibrated directly to give the dissipation factor at a particular frequency.

- The lower junction of the bridge is grounded. At the frequency normally used on this bridge, the reactances of capacitors C_3 and C_x are much higher than the resistances of R_1 and R_2. Hence if the junction of R_1 and R_2 is grounded, the detector is effectively at ground potential. This reduces any stray-capacitance effect, and makes the bridge more stable.

Power factor. The *power factor* (p.f.) of a series R.C. combination is defined as *the cosine of the phase angle of the circuit.*

$\therefore$ The power factor of the unknown, p.f. $= \dfrac{R_x}{Z_x}$

For phase angles very close to 90°, the reactance is almost equal to the impedance and we can approximate the power factor to,

$$\text{p.f.} \simeq \frac{R_x}{X_x} = \frac{R_x}{\frac{1}{\omega C_x}} = \omega R_x C_x \quad ...(6.72)$$

Dissipation factor (D). The *dissipation factor* of a series *R-C* circuit is defined *as the cotangent of the phase angle.*

i.e.,
$$D = \frac{R_x}{X_x} = \omega C_x R_x \quad ...(6.73)$$

Also, since the quality of a coil is defined by $Q = \frac{X_L}{R_L}$, we find that the dissipation factor, *D*, is the *reciprocal of the quality factor, Q,*

i.e.
$$D = \frac{1}{Q} \quad ...(6.74)$$

The dissipation factor tells us something about the quality of the capacitor; *i.e.*, how close the phase angle of the capacitor is to the ideal value of 90°.

Substituting the value of C_x in eqn. (6.71) and of R_x in eqn. (6.70) in the eqn. (6.74), we get

$$D = \omega \times \frac{R_1 C_3}{R_2} \times \frac{R_2 C_1}{C_3} = \omega R_1 C_1 \quad ...(6.75)$$

— When resistor R_1 in the Schering bridge of Fig. 6.34 has a fixed value, the dial of capacitor C_1 may be calibrated directly in dissipation factor *D*. This is the usual practice in a Schering bridge. The appearance of the term ω in the expression for the dissipation factor [eqn. (6.75)] means, of course, that the calibration of the C_1 dial holds for only one particular frequency at which dial is calibrated. A different frequency can be used, provided that a correction is made by multiplying the C_1 dial reading by the ratio of the two frequencies.

- Commercial units *measure from 100 pF–1μF, with ± 2% accuracy.*
- The Schering bridge is *widely used for testing small capacitors at low voltages with very high precision.*

... Fig. 6.34 has the following constants :

Solution. Refer to Fig. 6.45.

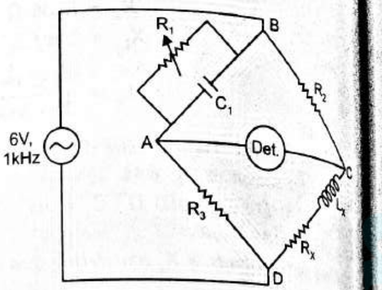


Fig. 6.45.

$$\bar{Z}_1 = R_1 \parallel X_{C_1}$$

or,
$$\bar{Y}_1 = \frac{1}{R_1} + j\omega C_1$$

$$= \frac{1}{600} + j(2\pi \times 1000 \times 1 \times 10^{-6})$$
$$= 1.67 \times 10^{-3} + j\, 6.283 \times 10^{-3}$$

$$\bar{Z}_2 = R_2 = 100\Omega$$
$$\bar{Z}_3 = R_3 = 1000\Omega$$
$$\bar{Z}_4 = \bar{Z}_x = R_x + jX_L$$

Using the basic balance equation, we get :

$$\bar{Z}_1 \bar{Z}_4 = \bar{Z}_2 \bar{Z}_3$$

or,
$$\bar{Z}_4 (= \bar{Z}_x) = \frac{\bar{Z}_2 \bar{Z}_3}{\bar{Z}_1} = \bar{Z}_2 \bar{Z}_3 \bar{Y}_1$$
$$= 100 \times 1000 \times [1.67 \times 10^{-3} + j\, 6.283 \times 10^{-3}]$$
$$= (167 + j\, 628.3)\ \Omega = (R_x + j\, X_L)\ \Omega$$

$\therefore$ $R_x = $ **167 Ω (Ans.)**

$$X_{L_x} = 628.3\ \Omega = 2\pi f L_x$$

$\therefore$
$$L_x = \frac{628.3}{2\pi \times 1000} \simeq 0.1 \text{ H (Ans.)}$$

6.3.11. Wien Bridge (*For measurement of* "Frequency")

The Wien bridge is *primarily known as a "frequency" determining bridge*. It can also be used for the *measurement of an unknown capacitor with great accuracy*.

Fig. 6.46 shows a Wien bridge. It has a series *R*-C combination in one arm and a parallel combination in the adjoining arm.

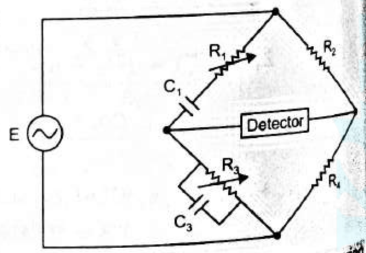


Fig. 6.46. Wien bridge for measuring frequency.

The impedance of one arm is,

$$\bar{Z}_1 = R_1 - \frac{j}{\omega C_1}$$

The admittance of parallel arm is

$$\bar{Y}_3 = \frac{1}{R_3} + j\omega C_3$$

Using the bridge balance equation, we have

$$\bar{Z}_1 \bar{Z}_4 = \bar{Z}_2 \bar{Z}_3$$

or,
$$\bar{Z}_1 \bar{Z}_4 = \bar{Z}_2 \times \frac{1}{\bar{Y}_3}$$

or,
$$\bar{Z}_2 = \bar{Z}_1 \bar{Z}_4 \bar{Y}_3$$

or,
$$R_2 = \left(R_1 - \frac{j}{\omega C_1}\right) R_4 \left(\frac{1}{R_3} + j\omega C_3\right) \quad ...(6.95)$$

Expanding this expression, we get

$$R_2 = \frac{R_1 R_4}{R_3} + j\omega R_1 R_4 C_3 - \frac{jR_4}{\omega C_1 R_3} + \frac{R_4 C_3}{C_1}$$

or,
$$R_2 = \left(\frac{R_1 R_4}{R_3} + \frac{R_4 C_3}{C_1}\right) - j\left(\frac{R_4}{\omega C_1 R_3} - \omega C_3 R_1 R_4\right) \quad ...(6.96)$$

Equating the *real terms*, we obtain

$$R_2 = \frac{R_1 R_4}{R_3} + \frac{R_4 C_3}{C_1}$$

or,
$$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{C_3}{C_1} \quad ...(6.97)$$

Equating the *imaginary terms*, we have

$$\omega C_3 R_1 R_4 = \frac{R_4}{\omega C_1 R_3} \quad ...(6.98)$$

or,
$$\omega^2 = \frac{1}{C_1 C_3 R_1 R_3} \quad ...[6.98\ (a)]$$

or,
$$\omega = \frac{1}{\sqrt{C_1 C_3 R_1 R_3}}$$

Since, $\omega = 2\pi f$, therefore,

$$2\pi f = \frac{1}{\sqrt{C_1 C_3 R_1 R_3}}$$

or,
$$f = \frac{1}{2\pi \sqrt{C_1 C_3 R_1 R_3}} \quad ...(6.99)$$

We observe that the two conditions for bridge balance, (eqns. 6.97, 6.99), result in an expression determining the required resistance ratio $\frac{R_2}{R_4}$, and another expression determining the frequency of the applied voltage. In other words, if we satisfy eqn. (6.97), and also excite the bridge with a frequency described by eqn. (6.99), the bridge will be in balance.

In most Wien bridge circuits, the components are chosen such that $R_1 = R_3$ and $C_1 = C_3$; this reduces eqn. (6.97) to $\frac{R_2}{R_4} = 2$ and eqn. (6.99) to

$$f = \frac{1}{2\pi RC} \quad ...(6.100)$$

which is *general expression* for the frequency of the Wien bridge.

– In a practical bridge, capacitors C_1 and C_3 are *fixed* capacitors, and resistors R_1 and R_3 are variable resistors *controlled by a common shaft*. Provided now that $R_2 = 2R_4$, the bridge may be used as a frequency-determining device balanced by a single *control. This control may be calibrated directly in terms of frequency.*

- By the use of this bridge an *accuracy of 0.5% - 1% can be easily obtained.*

- Because it is frequency sensitive, it is difficult to balance unless the waveform of the applied voltage is purely sinusoidal.

Applications :

1. The bridge is employed for *measuring frequency in the audio range.*

— The audio range is normally divided into 20Hz - 200Hz - 2kHz - 20kHz ranges. In this case, the resistances can be used for range changing and capacitors C_1 and C_3 for fine frequency control within the range.

2. The bridge can also be used for *measuring capacitances;* in this case operational frequency must be known.
3. The bridge is also used in a harmonic distortion analyser, as a notch filler, and in audio frequency and radio frequency oscillators as a frequency determining element.

Example 6.30. *Determine the equivalent parallel resistance and capacitance that causes a Wien bridge (Fig. 6.46) to null with the following component values :*

$R_1 = 2.8k\Omega$; $C_1 = 4.8\mu F$; $R_2 = 20k\Omega$; $R_4 = 80k\Omega$; $f = 2$ kHz;

Solution. *Given* : $f = 2\text{kHz}$ or $\omega = 2\pi f = 2\pi \times (2 \times 1000) = 12.57 \times 10^3$ rad/s.

$$R_3 = ?, C_3 = ?$$

We know that :
$$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{C_3}{C_1} \quad \text{...(Eqn. 6.97)}$$

and
$$\omega^2 = \frac{1}{C_1 C_3 R_1 R_3} \quad \text{...(Eqn. 6.98 (a))}$$

or,
$$C_3 = \frac{1}{w^2 C_1 R_1 R_3}$$

Substituting for C_3 in eqn. (6.97), we get

$$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{1}{\omega^2 C_1^2 R_1 R_3}$$

$$\frac{R_2}{R_4} = \frac{1}{R_3}\left(R_1 + \frac{1}{\omega^2 C_1^2 R_1}\right)$$

or,
$$R_3 = \frac{R_4}{R_2}\left(R_1 + \frac{1}{\omega^2 C_1^2 R_1}\right) \quad \text{...(i)}$$

$$= \frac{80\times10^3}{20\times10^3}\left[2.8\times10^3 + \frac{1}{(12.57\times10^3)^2 (4.8\times10^{-6})^2 (2.8\times10^3)}\right]$$

$$= 4\ (2.8 \times 10^3 + 0.098) = \mathbf{11.2\ k\Omega\ (Ans.)}$$

Now,
$$C_3 = \frac{1}{\omega^2 C_1 R_1 R_3} \quad \text{Rearranging (Eqn. 6.98a)}$$

$$= \frac{1}{(12.57\times10^3)^2 \times (4.8\times10^{-6}) \times (2.8\times10^3) \times 11.2\times10^3}$$

$$= \mathbf{42.04\ pF\ (Ans.)}$$

Calibration of Voltmeter

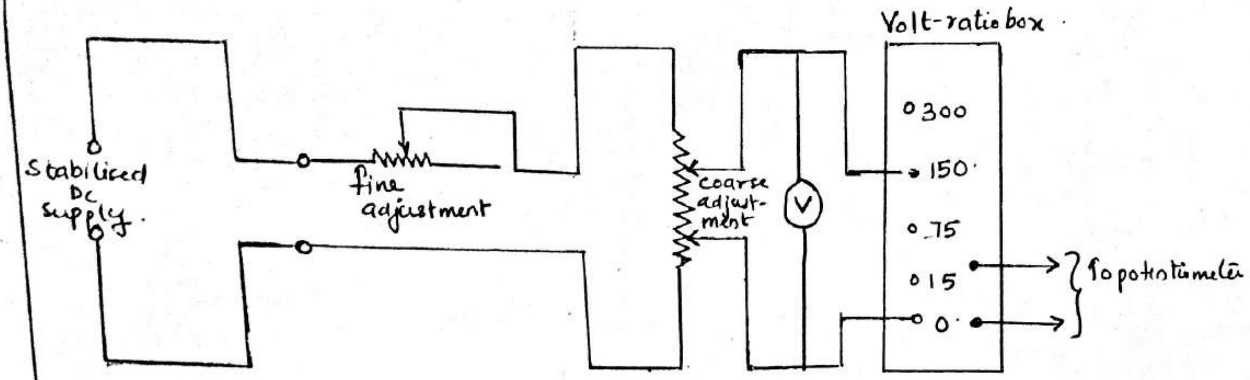


fig: Calibration of voltmeter with potentiometer

>> The primary requirement in this calibration process is that a suitable stable DC voltage supply is available otherwise any changes in the supply voltage will cause a corresponding change in the calibration of voltmeter.

>> The potential divider network consist of two rheostats, one for coarse & other for fine adjustments of calibrating voltage. These controls are connected to the supply source and with the help of these controls it is possible to adjust the voltage so that the pointer coincides exactly with a major division of voltmeter.

>> The voltage across the voltmeter is stepped down to a value suitable for application to a potentiometer with the help of a volt-ratio box.

>> The potentiometer measures the true value of voltage. If the reading of the potentiometer does not agree with the voltmeter reading, a -ve or +ve error is indicated.

Note:

⇒ Potentiometer is an instrument designed to measure an unknown voltage by comparing it with a known voltage. It has a high degree of accuracy.

Calibration of

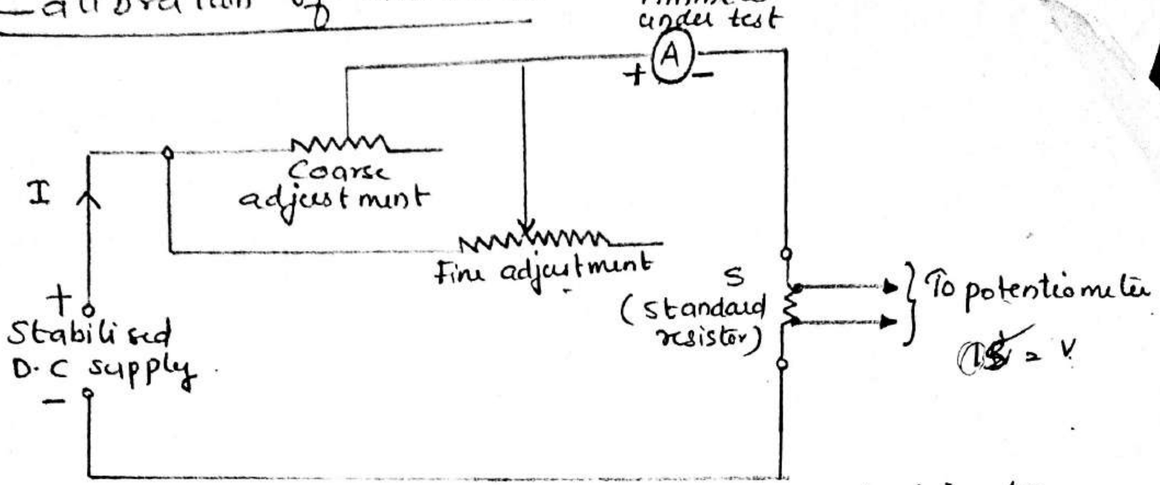


fig: calibration of an ammeter with potentiometer.

>> A standard resistor S of suitable value & sufficient current carrying capacity is placed in series with the ammeter under calibration.

>> The voltage across S is measured with the help of potentiometer and then the current through S can be computed, as given below.

$$\text{Current, } I = \frac{V_s}{S}$$ $V_s \rightarrow$ vtg across standard resistor S.

>> As the resistance of standard resistor S is accurately known the voltage across S is measured by a potentiometer, this method of calibrating an ammeter is very accurate.

• A calibration curve indicating the errors at various scale readings of the ammeter may be plotted.

Calibration of Wattmeter.

>> The current coil (cc) of wattmeter under calibration is supplied from low voltage supply & potential coil (PC) from the normal supply through potential divider.

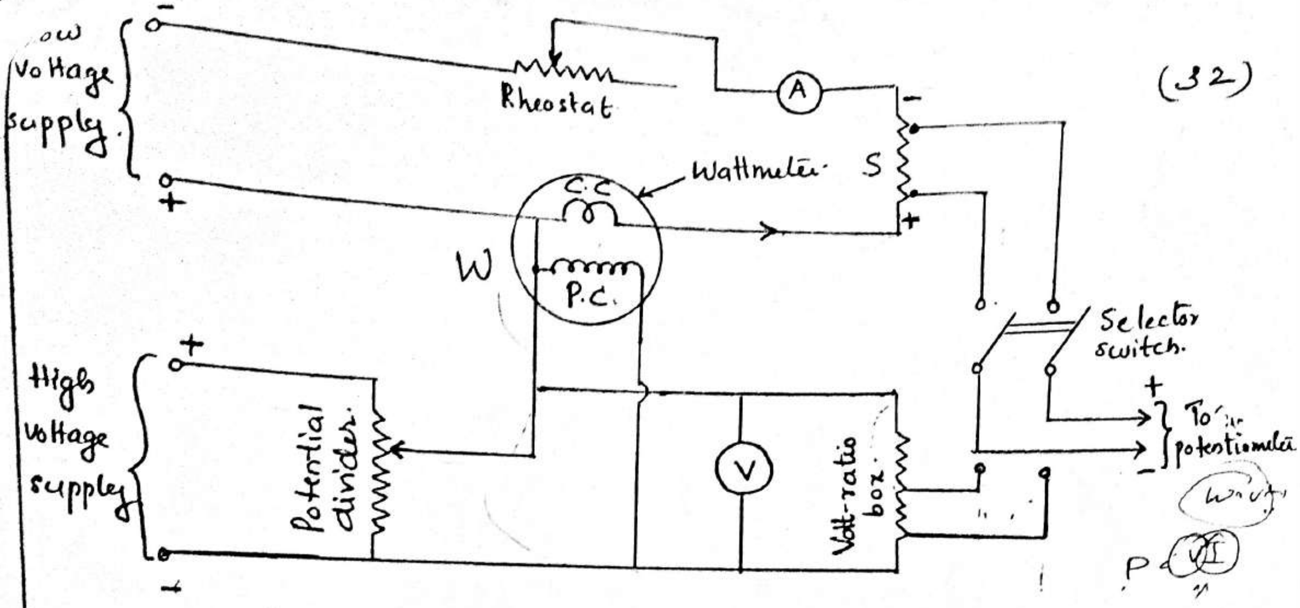


fig: Calibration of wattmeter with D.C potentiometer

>> The voltage V across the potential coil of the current I through current coil are measured in turn by the potentiometer employing a volt-ratio box and standard resistor, S as shown in fig.

>> The true power is then VI and the wattmeter reading may be compared with this value.

- This method of calibration of wattmeter is quite precise.

1. Electrostatic instruments or Electrostatic voltmeter

(5)

- These instruments may be used on both a.c & d.c
- They are particularly suited for high voltages.

>> The deflecting torque is produced by action of electric field on charged conductors, such instruments are essentially voltmeters but they may be used with the help of external components, to measure current & power.

>> Greatest use in the laboratory is for measurement of high voltages.

There are two ways in which the force act.

i. One type involves two oppositely charged electrodes one is fixed and other is movable due to force of attraction, the movable electrode is drawn towards the fixed one.

ii. In the other type, there are force of attraction or repulsion or both between the electrodes which cause rotary motion of the moving electrodes.

>> In both the cases the mechanism resembles a variable capacitor and the force or torque is due to the fact that mechanism tend to move the moving electrode to such a position where the energy stored is maximum.

>> These instruments may be used with the help of external components, to measure current & power.

Force & torque equations of electrostatic instrument

>> The stored energy is used as a basis for derivation of force & torque equations

Linear motion

>> Let a voltage V be applied between two parallel plates, one fixed (A) and the other movable (B)

>> The capacitance between the plates is C and the energy stored is

$$\frac{1}{2}CV^2$$

>> If the voltage is increased by dv, the force of attraction (F) will increase & the movable plate will move by a distance dx.

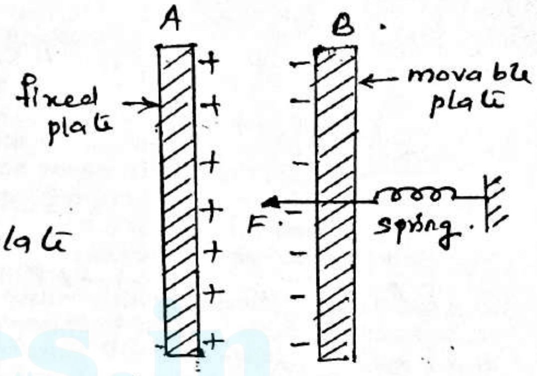


fig: Linear motion of electrostatic instrument

>> The capacitance current,

$$i = \frac{dQ}{dt} = \frac{d(CV)}{dt}$$

$$i = C\frac{dv}{dt} + V\frac{dc}{dt}$$

>> The energy input is; $Vi\,dt = V\left[C\frac{dv}{dt} + V\frac{dc}{dt}\right]dt$

$$Vi\,dt = V^2 dc + CV \cdot dv$$

>> The change in energy stored in electric field =

$$= \frac{1}{2}(C+dc)(V+dV)^2 - \frac{1}{2}CV^2$$

$$= \frac{1}{2}(C+dc)(V^2 + 2VdV + (dV)^2) - \frac{1}{2}CV^2$$

$$= \frac{1}{2}(CV^2 + CVdV + CdV^2 + V^2dc + 2VdVdc + dcV^2) - \frac{1}{2}CV^2$$

$$= \frac{1}{2}V^2dc + CV \cdot dV$$

neglecting the higher order terms as they are small quantities

3. From the principle of law of conservation of energy. (6)

input electrical energy = increase in stored energy + mechanical work done.

or $V^2 \cdot dc + CV \cdot dV = \frac{1}{2} V^2 dC + CV \cdot dV + F dx$.

Force, $F = \frac{1}{2} V^2 \frac{dc}{dx}$ Newton

Rotational Motion.

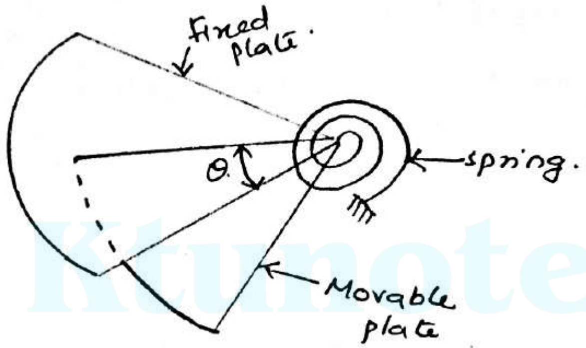


fig: rotational motion of electrostatic instrument.

If the motion is rotational the deflecting torque is given by. $T_d = \frac{1}{2} V^2 \frac{dc}{d\theta}$.

If the instrument is spring controlled

Controlling Torque $T_c = K\theta$.

K - spring constant
θ - deflection.

At steady deflection, $T_d = T_c$.

$$\frac{1}{2} V^2 \frac{dc}{d\theta} = K\theta.$$

$$\boxed{\text{deflection, } \theta = \frac{1}{2} \frac{V^2}{K} \cdot \frac{dc}{d\theta}}$$

6. Sphere g...

1. **Electrostatic voltmeter :**

The electrostatic voltmeters have been discussed in chapter 3. However, the design this type of meter has to be modified for high voltage use.

The following instruments may be used for measuring r.m.s. values of high voltages :

(i) Attracted disc voltmeters.

(ii) Attracted sphere voltmeter.

(iii) Oscillating ellipsoid voltmeter.

(i) ***Attracted disc voltmeters :***

Fig. 8.9 shows one form of these instruments :

- This voltmeter consists of two hollow metal mushroom shaped discs. One of the discs has a portion of its centre removed and a small movable disc enclosed. The movable disc is connected to the pointer of the instrument through a gear system. The other disc, which is fixed in position, is connected to the high voltage supply.
- The distance between the two discs remains *fixed for any measurement and can be changed for changing the range of the instrument.*

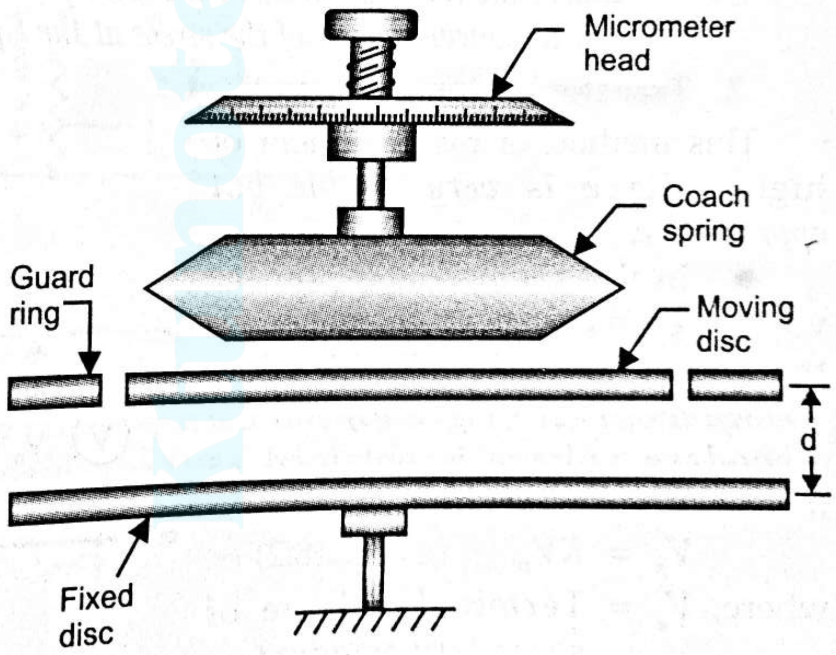


Fig. 8.9. Attracted disc type electrostatic voltmeter.

Sphere gaps.

The breakdown voltage of a sphere gap between 2 metallic sphere may be used for the measurement of voltages upto highest problem encountered in high voltage testing

In this method of measurement of extra high voltage two equal sized metal spheres made of brass, bronze, steel aluminium, Al alloy or copper separated by an (gas-gap) air gap are mounted vertically one above the other with the lower sphere earthed.

→ Voltage (Potential diff) between the spheres is raised till a spark passes between the 2 spheres. The value of voltage required to spark over depends upon the dielectric strength of air, the size of spheres, the distance between the spheres, humidity and temperature.

→ The spark gap over voltages are in terms of rms voltage. To obtain peak value of voltage, for sine wave. To obtain peak value of voltage, (rms value are multiplied by peak factor $\sqrt{2}$) and for checking & calibrating of voltmeters.

Advantages.

The method is cheap, simple & reliable.

1. The method is cheap, simple & reliable. Peak voltage of values ranging from 2KV to 2500KV
2. Peak voltage of values ranging from 2KV to 2500KV can be measured by this method also calibration of high voltage voltmeter can also done
3. sphere gap may also be used for measurement of voltage in surge tests

Disadvantages

1. This method doesnot provide a continuous record of voltage.

Precautions

1. Sphere should not be highly polished.
2. spheres should be properly cleaned to remove any extraneous matters.
3. Time interval between consecutive flash overs should be large enough to avoid appreciable heating of spheres.

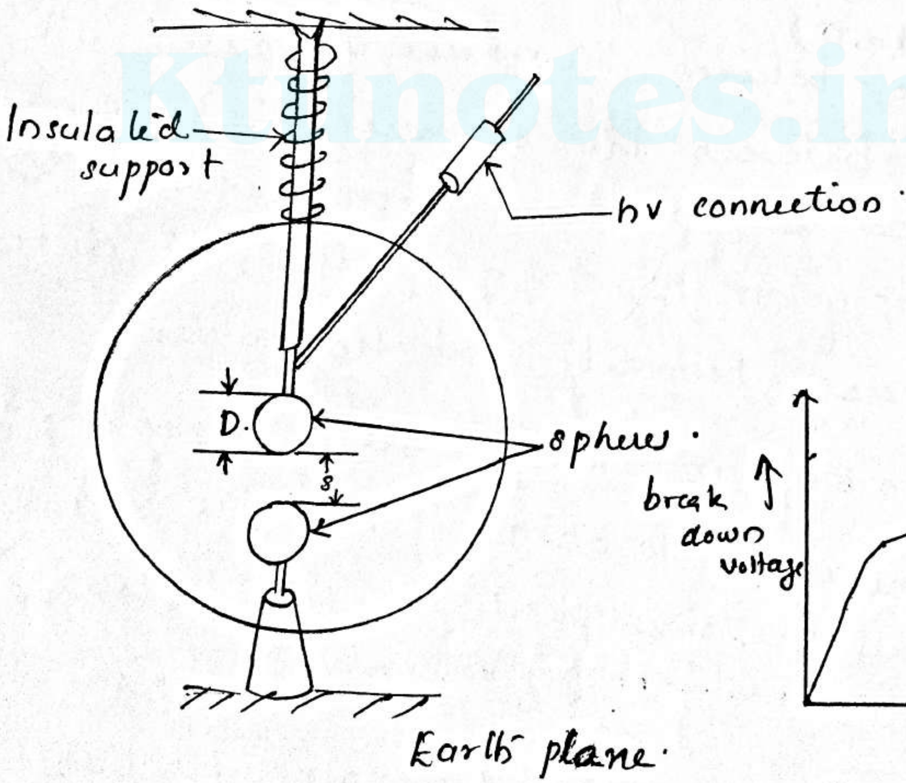


fig: Vertical arrangement.

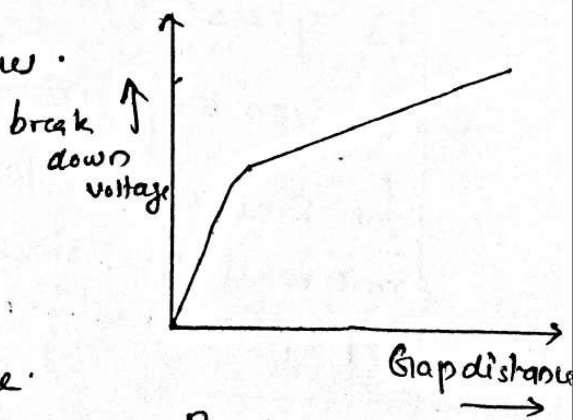


fig: Breakdown voltage characteristic of sphere gaps

Air density correction factor δ = ...